Type MS · Utility software: anti-malware, defrag and file tools

Answer 1 of 5 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

5(a)	Award **one** mark for each correct column: Translator Characteristic / Assembler / Interpreter / Compiler Does **not** have to be present in memory to execute the translated code. — ✓ — — ✓ Translates high level language source code. — — ✓ ✓ Executes high level code until either the program has finished, or a syntax error is encountered. — — ✓ — Translates low level language source code into machine code. — ✓ — —		3
5(b)	Award **two** marks for a linked explanation such as: • Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1) • Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1)	One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified. e.g. "A lower clock speed means processing will take longer" is generic for one mark. Low performance on its own is not enough.	2

16

Type MS · Utility software: anti-malware, defrag and file tools

Answer 1 of 5 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:
• Lossless (1)

Award one mark for any of the following up to a maximum of two marks:
• No data will be lost / the process is reversable (1)
• The manual will still be readable (1)
• Removal of data would change the meaning of text (1)
• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification

Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Utility software: anti-malware, defrag and file tools

Answer 1 of 5 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

5(e)	Award **one** mark for each of: Method 1 • (C AND NOT D) (1) • (NOT C AND D) (1) • <expr1> OR <expr2> (1) Method 2 • (C OR D) (1) • NOT (C AND D) (1) • <expr1> AND <expr2> (1) Example statements: (C AND NOT D) OR (NOT C AND D) (C OR D) AND NOT (C AND D)	Order of precedence is NOT, AND, OR Note other equivalent solutions MAX 2 if all terms correct apart from missing brackets that are required. Do not credit circuit diagrams.	3
		Total for question 5	**15**

Type MS · Utility software: anti-malware, defrag and file tools

Answer 1 of 5 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Utility software: anti-malware, defrag and file tools

Answer 2 of 5 · Question 3 part e · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part e in this mark scheme extract

Number		Guidance	
3(a)	Award **two** marks for an explanation that includes hardware **and** software. • The laptop is/contains the hardware (1) and the software allows the user to interact with it (1) • Hardware gives input and output devices (1) /memory, storage, cpu, processing ability • Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1) • Software is the set of instructions (1) that controls the hardware (1)		2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **two** marks from: • Lighter/thinner (1) • Less heat generated (1) • Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1) • Silent operation (1) • Less power required / longer battery life (1) • More robust (due to no moving parts) (1)		2
3(b)(ii)	Award **two** marks for a linked explanation such as: • Flash memory is non-volatile (1) whereas RAM is volatile (1) • Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1) • Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1)		2

Type MS · Utility software: anti-malware, defrag and file tools

Answer 2 of 5 · Question 3 part e · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked explanation such as: • Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1)		2

Question Number	Answer	Additional Guidance	Mark
3(d)	Award **one** mark for two correct rows. Award **two** marks for four correct rows.		2

Device	Input	Output	Both
Headset			✓
Mouse	✓		
Printer		✓	
Touch screen			✓

Question Number	Answer	Additional Guidance	Mark
3(e)	Award **two** marks from: • Antivirus (1) • Anti-spyware (1) • (Software) firewall (1)	Do not award tradenames	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 6 · find part b.i in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 6 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 3 of 6 · find part b.i in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 4 of 6 · find part b.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 5 of 6 · find part b.i in this mark scheme extract

Type MS · Utility software: anti-malware, defrag and file tools

Answer 3 of 5 · Question 6 part b.i · 2 marks

2021 NOV · Paper 01 · extract 6 of 6 · find part b.i in this mark scheme extract

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Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 6 · find part b.ii in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 6 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 3 of 6 · find part b.ii in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 4 of 6 · find part b.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 5 of 6 · find part b.ii in this mark scheme extract

Type MS · Utility software: anti-malware, defrag and file tools

Answer 4 of 5 · Question 6 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 6 of 6 · find part b.ii in this mark scheme extract

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Type MS · Utility software: anti-malware, defrag and file tools

Answer 5 of 5 · Question 3 part d · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

3(a)	Award up to **two** marks for a linked explanation, such as: • If the data is incomplete/inaccurate (1) the answers from the model might not be right (1) • If the assumptions the model is based on are inaccurate (1) the answers from the model might be incomplete/inaccurate (1)	Award one mark for identifying a problem and one mark for a reason There must be a clear cause and consequence/effect	2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for any of: • Pixels per inch/ppi (1) • Pixels per centimetre/ppc (1) • Dots per inch/dpi (1)	Ignore units cm^2 $inches^2$	1

Type MS · Utility software: anti-malware, defrag and file tools

Answer 5 of 5 · Question 3 part d · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	Award **two** marks for a linked description such as: • Each pixel/colour in the image (1) will require more bits to store (1)	Giving for formula file_size = width*height*colour_depth on its own is not enough	2

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked description such as: • System software is used to maintain/run the operation of the system/hardware (1) whereas application software allows the user to carry out a task (1)	For both marks both categories must be addressed Allow clear examples	2

Type MS · Utility software: anti-malware, defrag and file tools

Answer 5 of 5 · Question 3 part d · 2 marks

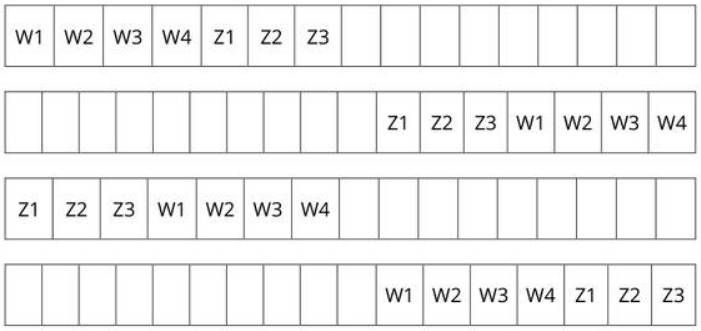
2023 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

<table>
<tr><th>Question number</th><th>Answer</th><th>Additional guidance</th><th>Mark</th></tr>
<tr><td>3(d)</td><td>Award one mark for each of:

• All file blocks of each file are in numeric sequence/one after another in numeric order (1)

• All blank space is together at the front/at the end (1)

Examples:

 
</td><td>Note: Order of letter and number does not matter. W1 or 1W are equally acceptable</td><td>2</td></tr>
</table>

Type MS · Utility software: anti-malware, defrag and file tools

Answer 5 of 5 · Question 3 part d · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)	Award up to **two** marks for a linked explanation such as: • The file size will not be decreased/will increase (1) because there are very few/no repeating patterns / because a run length of one would be added to each digit (1)		2

Question Number	Answer	Additional Guidance	Mark
3(f)	Award **one** mark for each correct cell:		4

Task	Translator
A guessing game that can be used on different computing platforms	Interpreter (1)
A screen driver for a new smartphone	Assembler (1)
A new version of a spreadsheet program for sale next year	Compiler (1)
Control software for an embedded system inside a new washing machine	Assembler (1)

Type MS · Utility software: anti-malware, defrag and file tools

Answer 5 of 5 · Question 3 part d · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark